

SULIT
SM015
Mathematics 1
Semester I
Session 2020/2021
2 hours

SM015
Matematik 1
Semester I
Sesi 2020/2021
2 jam



KEMENTERIAN PENDIDIKAN MALAYSIA

BAHAGIAN MATRIKULASI
MATRICULATION DIVISION

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Kertas soalan ini mengandungi **13** halaman bercetak.

This question paper consists of 13 printed pages.

PUSAT SUMBER
KOLEJ MATRIKULASI PAHANG

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INSTRUCTIONS TO CANDIDATE:

This question paper consists of **Section A** and **Section B**.

Section A consists of 3 questions. **Section B** consists of 7 questions.

Answer **all** questions.

All answers must be written in the answer booklet provided. Use a new page for each question.

Marks for each question or section are shown in the bracket at the end of the question or section.

All steps must be shown clearly.

Only non-programmable scientific calculators can be used.

Numerical answers may be given in the form of π , e , surd, simplest fractions or up to three significant figures, where appropriate, unless stated otherwise in the question.

ARAHAN KEPADA CALON:

*Kertas soalan ini mengandungi **Bahagian A** dan **Bahagian B**.*

***Bahagian A** mengandungi 3 soalan. **Bahagian B** mengandungi 7 soalan.*

*Jawab **semua** soalan.*

Semua jawapan hendaklah ditulis pada buku jawapan yang disediakan. Gunakan muka surat baharu bagi nombor soalan yang berbeza.

Markah yang diperuntukkan bagi setiap soalan atau bahagian soalan ditunjukkan dalam kurungan pada penghujung soalan atau bahagian soalan.

Semua langkah kerja hendaklah ditunjukkan dengan jelas.

Kalkulator saintifik yang tidak boleh diprogramkan sahaja yang boleh digunakan.

Jawapan berangka boleh diberi dalam bentuk π , e , surd, pecahan termudah atau sehingga tiga angka bererti, di mana-mana yang sesuai, kecuali jika dinyatakan dalam soalan.

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Trigonometry
Trigonometri

$$\sin (A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\begin{aligned}\cos 2A &= \cos^2 A - \sin^2 A \\ &= 2 \cos^2 A - 1 \\ &= 1 - 2 \sin^2 A\end{aligned}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Quadratic equation $ax^2 + bx + c = 0$:

Persamaan kuadratik $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Differentiation

Pembezaan

$f(x)$

$f'(x)$

$\cot x$

$-\operatorname{cosec}^2 x$

$\sec x$

$\sec x \tan x$

$\operatorname{cosec} x$

$-\operatorname{cosec} x \cot x$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2 y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

Sphere

Sfera

$$V = \frac{4}{3} \pi r^3$$

$$S = 4 \pi r^2$$

Right circular cone

Kon membulat tegak

$$V = \frac{1}{3} \pi r^2 h$$

$$S = \pi r^2 + \pi r l$$

Right circular cylinder

Silinder membulat tegak

$$V = \pi r^2 h$$

$$S = 2\pi r^2 + 2\pi r h$$

SECTION A [25 marks]

BAHAGIAN A [25 markah]

This section consists of 3 questions. Answer **all** questions.
*Bahagian ini mengandungi 3 soalan. Jawab **semua** soalan.*

- 1 (a) Given the function

Diberi fungsi

$$f(x) = \begin{cases} x+1, & x < 1 \\ \frac{x^2 - x - a}{x - a}, & x \geq 1, x \neq a \end{cases}$$

is continuous at $x=1$. Find the value of a . Hence, determine $\lim_{x \rightarrow a} f(x)$.

adalah selanjutnya pada $x=1$. Cari nilai a . Seterusnya, tentukan $\lim_{x \rightarrow a} f(x)$.

[6 marks]

[6 markah]

- (b) Evaluate $\lim_{x \rightarrow +\infty} \left(\frac{2x+5}{3x-4} \right)^{\frac{1}{3}}$.

Nilaikan $\lim_{x \rightarrow +\infty} \left(\frac{2x+5}{3x-4} \right)^{\frac{1}{3}}$.

[3 marks]

[3 markah]

- 2 (a) Given the parametric equation

Diberi persamaan berparameter

$$x = \frac{1}{t} + 3t^2, \quad y = t^2 - 4\sqrt{t}, \quad t > 0.$$

Find $\frac{d^2y}{dx^2}$ in terms of t . Hence, evaluate $\frac{d^2y}{dx^2}$ at $t = 1$.

Cari $\frac{d^2y}{dx^2}$ dalam sebutan t . Seterusnya, nilaikan $\frac{d^2y}{dx^2}$ pada $t = 1$.

[6 marks]

[6 markah]

- (b) Given $y = \ln\left(\frac{\sin 2x}{1 - \cos 2x}\right)$. Show that $\frac{dy}{dx} = -2 \operatorname{cosec} 2x$.

Diberi $y = \ln\left(\frac{\sin 2x}{1 - \cos 2x}\right)$. Tunjukkan bahawa $\frac{dy}{dx} = -2 \operatorname{cosec} 2x$.

[5 marks]

[5 markah]

- 3 A farmer has a 100 meter of fencing wants to fence off a rectangular piece of land that borders a straight river. The part of the land which borders the river needs no fencing. Find the dimensions of the enclosed region that has the maximum area.

Seorang petani mempunyai 100 meter pagar untuk memagari sebidang tanah berbentuk segi empat tepat yang bersempadan dengan sungai yang lurus. Bahagian tanah yang bersempadan dengan sungai tidak perlu dipagar. Cari dimensi kawasan berpagar yang menghasilkan luas maksimum.

[5 marks]

[5 markah]

SECTION B [75 marks]

BAHAGIAN B [75 markah]

This section consists of 7 questions. Answer **all** questions.
*Bahagian ini mengandungi 7 soalan. Jawab **semua** soalan.*

- 1 (a) Solve the equation

Selesaikan persamaan

$$\ln 2^{4x-1} = \ln 8^{x+5} + \log_2 16^{1-2x}$$

by expressing your answer in terms of $\ln 2$.

dengan mengungkapkan jawapan anda dalam sebutan $\ln 2$.

[4 marks]

[4 markah]

- (b) Given $z_1 = 2 - 3i$ and $z_2 = 4 + 2i$.

Without using calculator, find $z_3 = \frac{\bar{z}_1}{z_2}$ and hence, evaluate $|z_3|$.

Diberi $z_1 = 2 - 3i$ dan $z_2 = 4 + 2i$.

Tanpa menggunakan kalkulator, cari $z_3 = \frac{\bar{z}_1}{z_2}$ dan seterusnya nilaikan $|z_3|$.

[5 marks]

[5 markah]

- 2 Determine the solution set of the following inequalities.

Tentukan set penyelesaian bagi ketaksamaan berikut.

(a) $\frac{4x}{4-x} \geq 6.$

[5 marks]

[5 markah]

(b) $|x-6| < 3|x-2|.$

[6 marks]

[6 markah]

- 3 FIGURE 3 shows a function $f(x)$ with domain $-2 \leq x \leq 5$.

RAJAH 3 menunjukkan suatu fungsi $f(x)$ dengan domain $-2 \leq x \leq 5$.

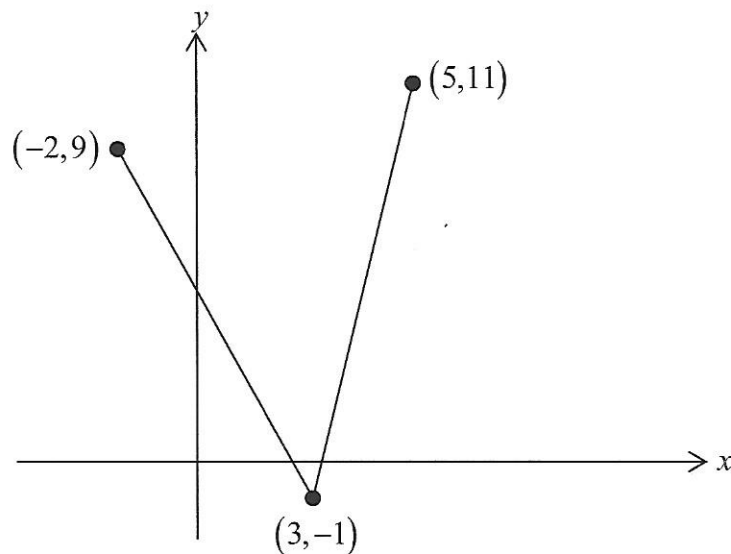


FIGURE 3

RAJAH 3

- (a) State the range of $f(x)$.

Nyatakan julat bagi $f(x)$.

[1 mark]

[1 markah]

- (b) Determine the function of $f(x)$.

Tentukan fungsi $f(x)$.

[4 marks]

[4 markah]

- (c) Evaluate $(f \circ f)(0)$.

Nilaikan $(f \circ f)(0)$.

[2 marks]

[2 markah]

- 4 Given the function $f(x) = \frac{3}{x^2 - 4}$, $x \neq -2, 2$ and $(f \circ g)(x) = \frac{3}{x - 5}$, $x \neq 5$.

Diberi fungsi $f(x) = \frac{3}{x^2 - 4}$, $x \neq -2, 2$ dan $(f \circ g)(x) = \frac{3}{x - 5}$, $x \neq 5$.

- (a) Obtain $g(x) \geq 0$ and state its domain.

Dapatkan $g(x) \geq 0$ dan nyatakan domainnya.

[4 marks]

[4 markah]

- (b) Show that $g^{-1}(x)$ exists. Hence, find $g^{-1}(x)$ and its domain.

Tunjukkan $g^{-1}(x)$ wujud. Seterusnya, cari $g^{-1}(x)$ dan domainnya.

[5 marks]

[5 markah]

- (c) Find the values of a and b if $h(x) = ax + b$, such that $(h \circ g^{-1})(x) = 5x^2 + 12$.

Cari nilai a dan b jika $h(x) = ax + b$, supaya $(h \circ g^{-1})(x) = 5x^2 + 12$.

[4 marks]

[4 markah]

5 Given

Diberi

$$f(x) = \frac{2x^2 + 2}{x^2 - 4}.$$

- (a) Find the vertical and horizontal asymptotes of $f(x)$.

Cari asimptot mencancang dan mengufuk bagi $f(x)$.

[8 marks]

[8 markah]

- (b) Hence, sketch the graph $f(x)$.

Seterusnya, lakar graf $f(x)$.

[3 marks]

[3 markah]

- 6 (a) Given $y = \tan\left(\frac{\pi}{4} + x\right)$, show that $\frac{d^2y}{dx^2} - 2y\frac{dy}{dx} = 0$.

Diberi $y = \tan\left(\frac{\pi}{4} + x\right)$, tunjukkan bahawa $\frac{d^2y}{dx^2} - 2y\frac{dy}{dx} = 0$.

[5 marks]

[5 markah]

- (b) Differentiate $g(x) = \frac{e^x}{e^x - 2}$, $x \neq \ln 2$ with respect to x .

Hence, calculate the exact value of x if $g'(x) = -2$.

Bezakan $g(x) = \frac{e^x}{e^x - 2}$, $x \neq \ln 2$ terhadap x .

Seterusnya, hitung nilai tepat bagi x jika $g'(x) = -2$.

[6 marks]

[6 markah]

- (c) Find the value of tangent to the curve $y^2 = x^3$ and $2x^2 + 3y^2 = k$, where k is a constant, at the points $(1, -1)$ and $(1, 1)$.

Hence, state the relationship between the tangent values at these two points.

Cari nilai tangen kepada lengkung $y^2 = x^3$ dan $2x^2 + 3y^2 = k$, dengan k adalah pemalar, pada titik $(1, -1)$ dan $(1, 1)$.

Seterusnya, nyatakan hubungan antara nilai tangen pada dua titik tersebut.

[5 marks]

[5 markah]

- 7 Consider a cone as shown in **FIGURE 7**.

Pertimbangkan satu kon seperti yang ditunjukkan dalam RAJAH 7.

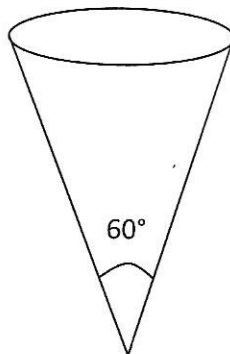


FIGURE 7

RAJAH 7

Water flows out from a cone through a hole located at the bottom tip at the rate of $5\text{cm}^3/\text{s}$. Find the rate of change of the height of the water when the volume of water in the cone is $3000\pi\text{ cm}^3$.

Air mengalir keluar daripada kon melalui lubang yang terletak di bucu bawahnya pada kadar $5\text{cm}^3/\text{s}$. Cari kadar perubahan ketinggian air apabila isipadu air dalam kon ialah $3000\pi\text{ cm}^3$.

[8 marks]

[8 markah]

END OF QUESTION PAPER

KERTAS SOALAN TAMAT